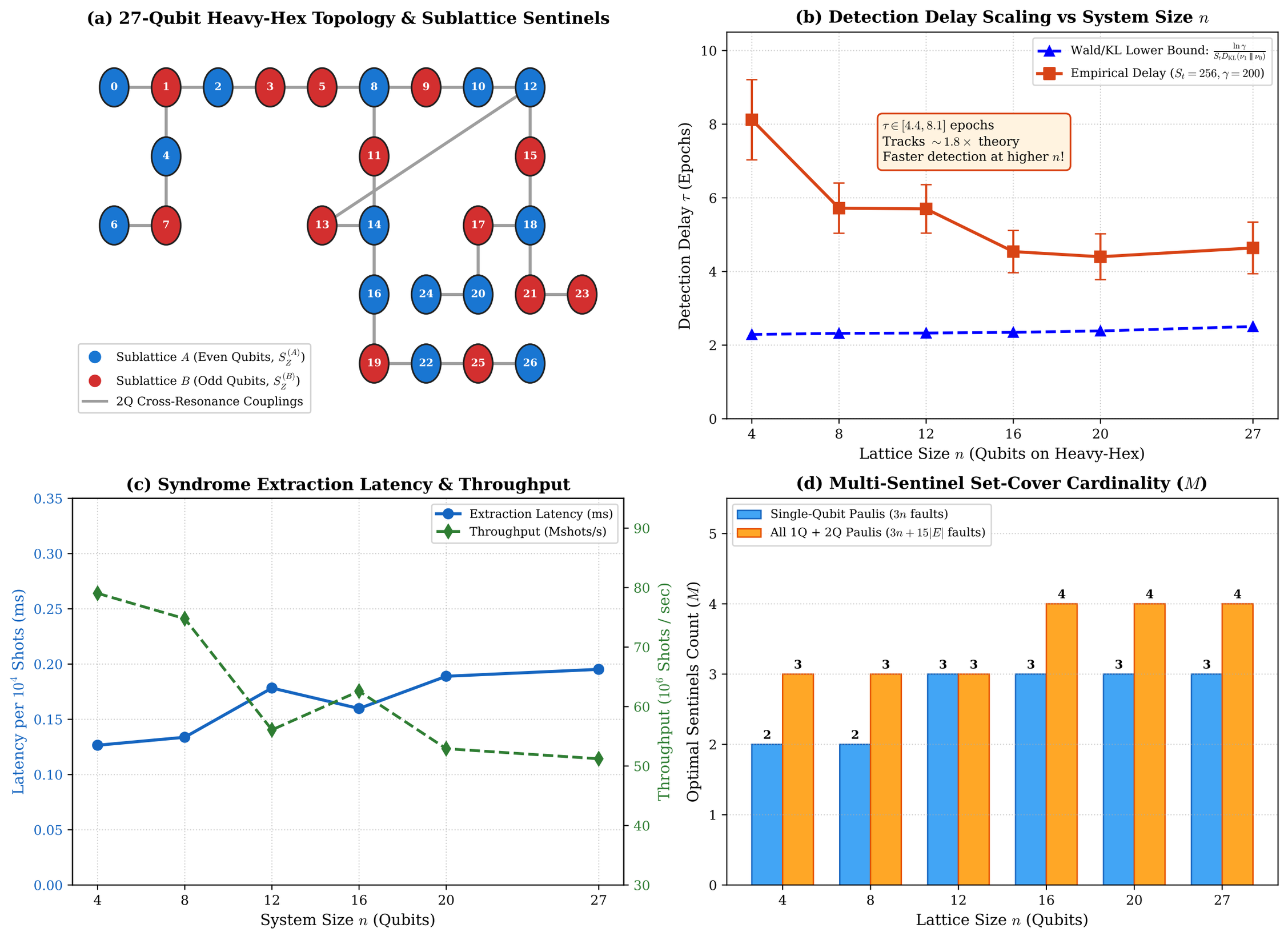

(a) 27-Qubit Heavy-Hex Topology & Sublattice Sentinels
0
1
2
3
5
8
9
10
12
4
11
15
6
7
13
14
17
18
16
24
20
21
23
19
22
25
26
Sublattice A (Even Qubits, $S_Z^{(A)}$)
Sublattice B (Odd Qubits, $S_Z^{(B)}$)
2Q Cross-Resonance Couplings
(b) Detection Delay Scaling vs System Size n
Wald/KL Lower Bound: $\frac{\ln \gamma}{S_t D_{\mathrm{KL}}(\nu_1 \| \nu_0)}$
Empirical Delay ($S_t = 256, \gamma = 200$)
$\tau \in [4.4, 8.1]$ epochs
Tracks $\sim 1.8\times$ theory
Faster detection at higher n!
Detection Delay τ (Epochs)
Lattice Size n (Qubits on Heavy-Hex)
(c) Syndrome Extraction Latency & Throughput
Extraction Latency (ms)
Throughput (Mshots/s)
Latency per 10^4 Shots (ms)
Throughput (10^6 Shots / sec)
System Size n (Qubits)
(d) Multi-Sentinel Set-Cover Cardinality (M)
Single-Qubit Paulis ($3n$ faults)
All 1Q + 2Q Paulis ($3n + 15|E|$ faults)
Optimal Sentinels Count (M)
Lattice Size n (Qubits)